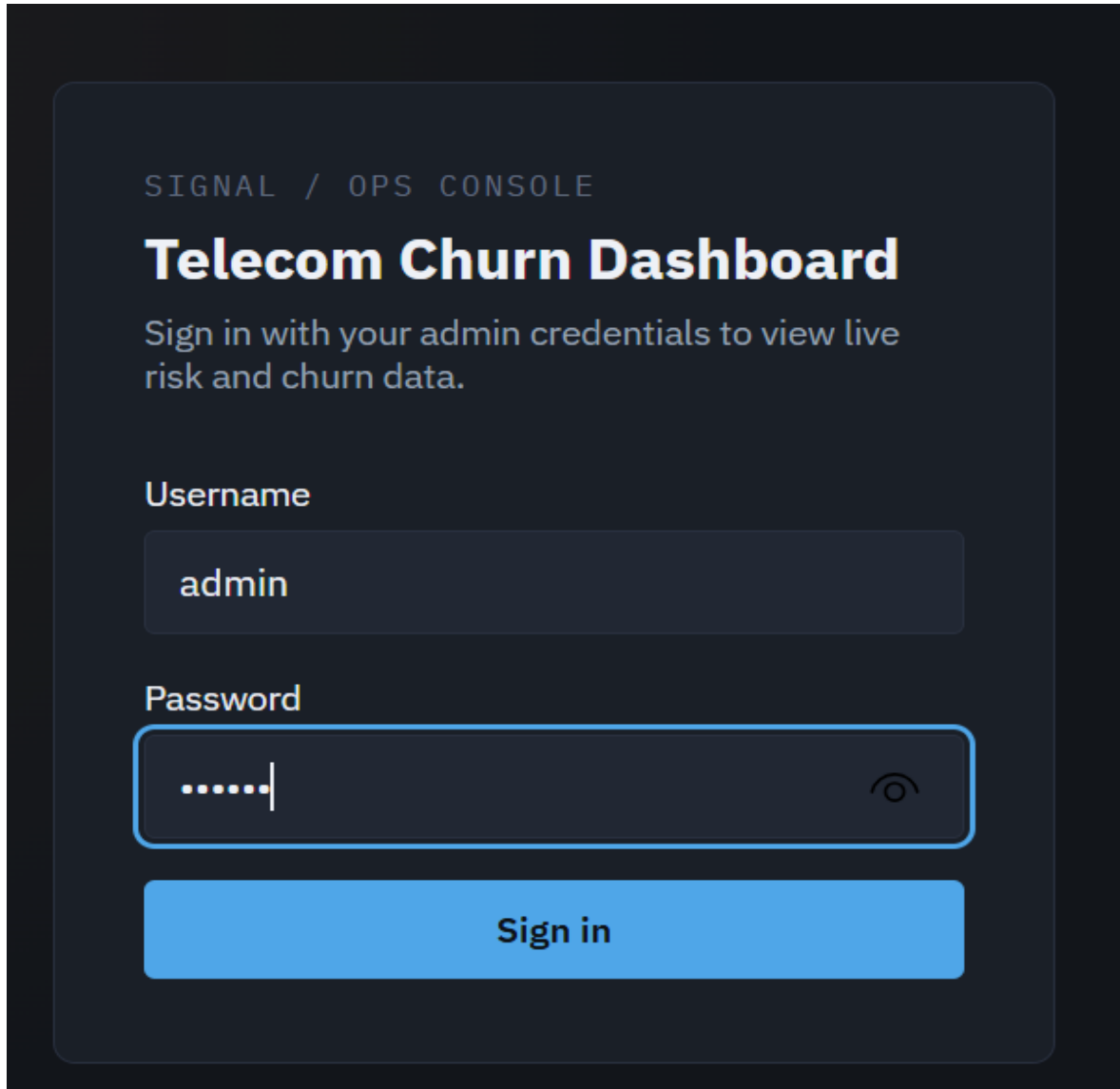


CUSTOMER RETENTION INTELLIGENCE SYSTEM



The image shows a dark-themed sign-in form for the 'Telecom Churn Dashboard'. At the top, it says 'SIGNAL / OPS CONSOLE' in a light blue font. Below that is the title 'Telecom Churn Dashboard' in a large, bold, white font. A subtitle in a smaller white font reads: 'Sign in with your admin credentials to view live risk and churn data.' The form contains two input fields: 'Username' with the value 'admin' and 'Password' with masked characters '.....'. A blue 'Sign in' button is at the bottom. A red rectangle highlights the password field and the button.

SIGNAL / OPS CONSOLE

Telecom Churn Dashboard

Sign in with your admin credentials to view live risk and churn data.

Username

admin

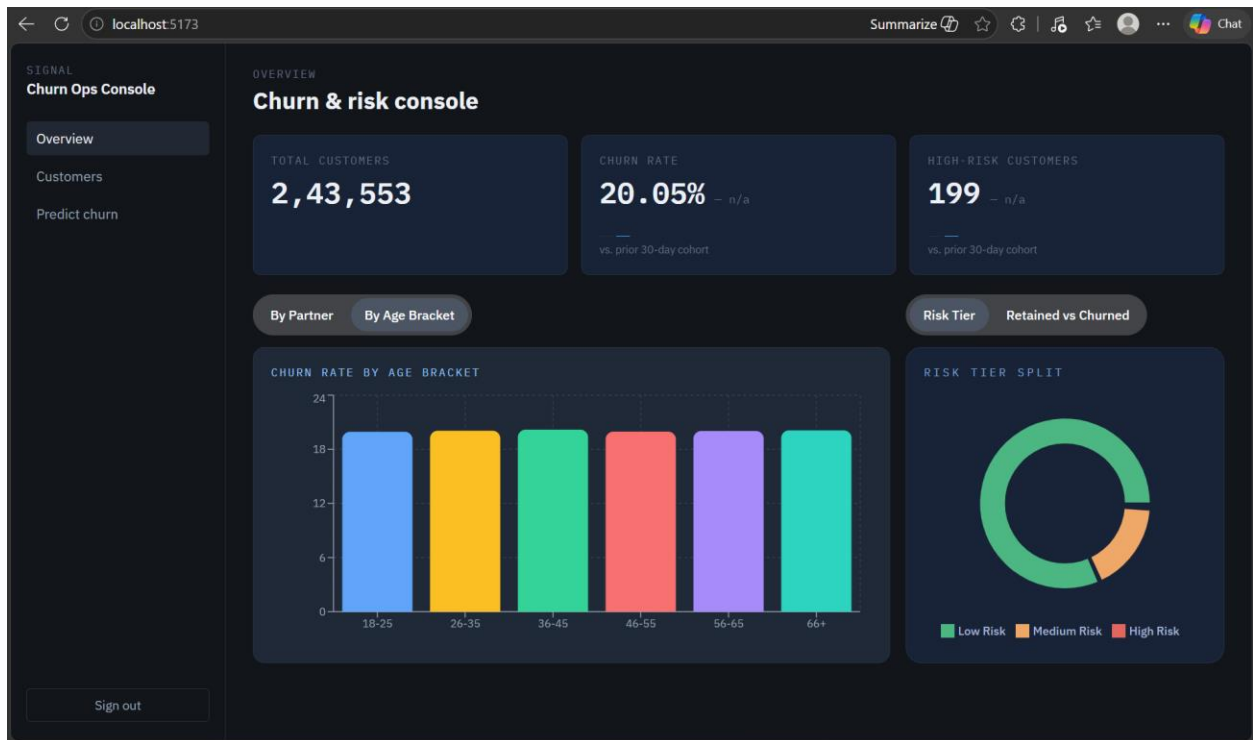
Password

.....

Sign in

Page 1 — Authentication (Sign In)

Every route in the Phase 3 FastAPI layer except `/health` requires OAuth2 bearer authentication. This sign-in screen posts the admin credentials to `POST /token`, which returns a bearer token; an Axios interceptor in `AuthContext` then attaches that token to every subsequent dashboard request.



Page 2 — Overview: Churn & Risk Console

The Overview tab is Phase 4's KPI-card row (total customers, churn rate, high-risk count, with trend deltas) plus the churn-by-age-bracket bar chart and risk-tier-split donut chart, drawn from GET /dashboard/kpis and the /dashboard/analytics endpoints. The Low/Medium/High tiers shown here come from the 7-point risk score calculated once at insertion time in Phase 2 and stored on each customer row, rather than recomputed per request.

← ↻ 📄 localhost:5173 Summarize 📄 ⚙️ 🎵 ⚙️ 👤 ⋮ 🗨 Chat

SIGNAL

Churn Ops Console

Overview

Customers

Predict churn

OVERVIEW

Churn & risk console

📊 Low Risk 📊 Medium Risk 📊 High Risk Hide filters ▲

PARTNER

Reliance Jio

Vodafone

BSNL

Airtel

STATE

Karnataka

Mizoram

Arunachal Pradesh

Tamil Nadu

Tripura

Uttarakhand

TENURE (DAYS)

min — max

AGE

min — max

SALARY (\$)

min — max

ID	RISK	PARTNER	CITY	STATE	AGE	TENURE (D)	SALARY	CHURNED
1	📊 Low Risk	Reliance Jio	Kolkata	Karnataka	25	1219	\$1,24,962	No
2	📊 Low Risk	Reliance Jio	Mumbai	Mizoram	55	1219	\$1,30,556	No
3	📊 Low Risk	Vodafone	Delhi	Arunachal Pradesh	57	1219	\$1,48,828	Yes
4	📊 Low Risk	BSNL	Kolkata	Tamil Nadu	46	1219	\$38,722	Yes
5	📊 Low Risk	BSNL	Delhi	Tripura	26	1219	\$55,098	No
6	📊 Low Risk	Vodafone	Chennai	Uttarakhand	36	1219	\$73,452	No
7	📊 Low Risk	BSNL	Delhi	Karnataka	60	1219	\$1,10,035	No

Page 3 — Customer Explorer: Filters

This view implements Phase 4's combined filter menu — partner, state, city, risk tier, and tenure/salary/age ranges, plus free-text search by ID or city — layered over the paginated, filterable GET `/dashboard/customers` endpoint from the Phase 3 dashboard router.

Churn & risk console

Search by ID or city...

Low Risk Medium Risk High Risk More filters

ID	RISK	PARTNER	CITY	STATE	AGE	TENURE (D)	SALARY	CHURNED
1	Low Risk	Reliance Jio	Kolkata	Karnataka	25	1219	\$1,24,962	No
2	Low Risk	Reliance Jio	Mumbai	Mizoram	55	1219	\$1,30,556	No
3	Low Risk	Vodafone	Delhi	Arunachal Pradesh	57	1219	\$1,48,828	Yes
4	Low Risk	BSNL	Kolkata	Tamil Nadu	46	1219	\$38,722	Yes
5	Low Risk	BSNL	Delhi	Tripura	26	1219	\$55,098	No
6	Low Risk	Vodafone	Chennai	Uttarakhand	36	1219	\$73,452	No
7	Low Risk	BSNL	Delhi	Karnataka	60	1219	\$1,10,035	No
8	Low Risk	BSNL	Kolkata	Arunachal Pradesh	46	1219	\$1,04,541	Yes
9	Low Risk	Reliance Jio	Mumbai	Himachal Pradesh	53	1219	\$79,439	No
10	Low Risk	BSNL	Mumbai	Rajasthan	57	1219	\$1,26,422	No
11	Medium Risk	Airtel	Chennai	Uttarakhand	44	1219	\$1,33,288	Yes
12	Low Risk	Airtel	Mumbai	Odisha	56	1219	\$1,35,129	No
13	Low Risk	Vodafone	Hyderabad	Uttar Pradesh	64	1219	\$1,36,967	No

Page 4 — Customer Explorer: Table

With the filter panel collapsed, the same GET /dashboard/customers endpoint backs Phase 4's paginated, sortable customer table, listing each customer's risk tier, partner, city, state, age, tenure, salary, and churn outcome.

ID	Risk	Carrier	City	State
4	Low Risk	BSNL	Kolkata	Tamil Nadu
5	Low Risk	BSNL	Delhi	Tripura
6	Low Risk	Vodafone	Chennai	Uttarakhand
7	Low Risk	BSNL	Delhi	Karnataka
8	Low Risk	BSNL	Kolkata	Arunachal Pradesh
9	Low Risk	Reliance Jio	Mumbai	Himachal Pradesh
10	Low Risk	BSNL	Mumbai	Rajasthan
11	Medium Risk	Airtel	Chennai	Uttarakhand
12	Low Risk	Airtel	Mumbai	Odisha
13	Low Risk	Vodafone	Hyderabad	Uttar Pradesh
14	Low Risk	Vodafone	Kolkata	Chhattisgarh
15	Low Risk	Vodafone	Delhi	Uttar Pradesh
16	Low Risk	BSNL	Mumbai	Tamil Nadu
17	Low Risk	Vodafone	Hyderabad	Himachal Pradesh
18	Low Risk	Reliance Jio	Hyderabad	Madhya Pradesh
19	Low Risk	Airtel	Delhi	Manipur
20	Medium Risk	Vodafone	Hyderabad	Uttar Pradesh
21	Low Risk	Vodafone	Bangalore	Goa
22	Low Risk	Vodafone	Bangalore	West Bengal
23	Medium Risk	BSNL	Chennai	Arunachal Pradesh

CUSTOMER DETAIL

#13

Low Risk

Risk score 1 / 7

PROFILE

Gender

Female

Age

64

Registered

2020-01-01

Tenure

1219 days

Dependents

4

Est. salary

\$1,36,967

Churned

No

ACCOUNT

Partner

Vodafone

Location

Hyderabad, Uttar Pradesh (339770)

USAGE

Calls made

40

SMS sent

4

Data used

1911 GB

RISK BREAKDOWN

○ Tenure < 180 days

0

○ Age between 18-30

0

○ Dependents <= 1

0

○ Calls made < 10

0

● SMS sent < 20

+1

○ Data used < 1 GB

0

○ High salary + low usage (income paradox)

0

Run churn prediction for this customer →

Page 5 — Customer Detail Drawer

Selecting a row opens the customer detail drawer sourced from GET `/dashboard/customers/{id}/detail`: profile and account info, the usage breakdown, and the per-factor risk-score triggers defined in Phase 2 (tenure < 180 days, age 18–30, ≤ 1 dependent, calls made < 10, SMS sent < 20, data used < 1 GB, and the high-salary/low-usage "income paradox"). The "Run churn prediction for this customer" button hands off to the Prediction tab pre-filled with this customer's stored attributes.

SIGNAL

Churn Ops Console

Overview

Customers

Predict churn

Sign out

ML PREDICTION

Churn prediction

Runs logistic regression, random forest, and xgboost together — the result is a majority vote across all three.

CUSTOMER FEATURES

Customer ID (optional)

13

Age *

64

Gender

Female

▼

Tenure (days)

1219

Dependents *

4

Estimated salary (\$) *

136967

Calls made *

40

SMS sent *

4

Data used (GB) *

1911

Telecom partner *

Vodafone

Pincode (optional)

339770

Predict churn

Reset

PREDICTION RESULT

Customer #13

38%

38% churn probability

Medium risk

Prediction: Will stay · Model agreement: 67%

MODEL VOTES

logistic regression

Churn

random forest

Stay

xgboost

Stay

RECOMMENDATION

Monitor customer engagement and consider retention offers.

Page 6 — Churn Prediction Workbench

Added in Phase 6, this workbench's PredictionForm posts the customer feature set to POST /ml/predict-churn, which runs a majority-vote ensemble of logistic regression, random forest, and XGBoost. PredictionResult then surfaces the churn probability, risk level, recommendation, and the per-model vote breakdown — exposed via an explicit model_votes field added to ChurnPredictionOutput after FastAPI's response_model was found to silently strip undeclared fields.